1.17 Proportion

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4: Negative and Fractional Indices

17

$a^0 = 1$
$a^{\frac{1}{m}} = \sqrt[m]{a}$
$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$
$a^{-m} = \frac{1}{a^m}$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3.
Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $27^{-\frac{1}{3}}$

Evaluate $32^{\frac{2}{5}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

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Writing a surd using indices

18

If $9\sqrt{3} = 3^k$, determine the value of k.

The key here is to write everything as powers with a consistent base, in this case, 3.

Note: In algebra we like to avoid mixed numbers. So don't write $3^{2\frac{1}{2}}$

1.17 Proportion

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Further Examples

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[Edexcel IGCSE May14-4H]
Given that

$$\left(2^{\frac{1}{2}}\right)^{n} = \frac{2^{x}}{8^{y}}$$

Express n in terms of x and y.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^{x}$, express the following in terms of y.

$$\frac{1}{4^{2x-3}}$$

Write your expression in its simplest form.

Write 8 as a power of 2 (putting a bracket around it), for consistency of base with the other powers.

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Test Your Understanding

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Edexcel Paper 2 – May 2019

Given $2^{x} \times 4^{y} = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

1.17 Proportion

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Exercise 1D

21

4 a Find the value of $81^{\frac{1}{4}}$. **b** Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants. **a** $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge

[MAT 2007 1A]

Let r and s be integers. Then

$$\frac{6^{r+s} \times 12^{r-s}}{8^r \times 9^{r+2s}}$$

is an integer if

- $r + s \leq 0$
- $s \leq 0$
- $r \leq 0$
- $r \geq s$

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GCSE RECAP: Reciprocal Graphs

21

Sketch $y = \frac{1}{x}$



Sketch $y = -\frac{3}{x}$

Note: The scaling caused by the 3 isn't observable for this graph in isolation because the axes have no scale. This will only be observation for multiple graphs on the same axes.

1.17 Proportion

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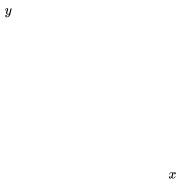
Reciprocal Graphs

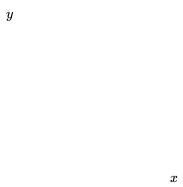
22

Sketch $y = \frac{3}{x^2}$

This is new to the A Level 2017 syllabus.

Sketch $y = -\frac{4}{x^2}$





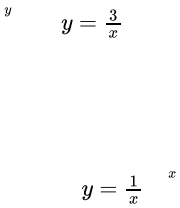
Hint: Note that anything squared will always be at least 0.

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Reciprocal Graphs

23

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$



The y value for $y = \frac{3}{x}$ will be 3 times greater than $y = \frac{1}{x}$

1.17 Proportion

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Exercise 4C – Pure Y1 page 67

24

1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$

d $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

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Exercise 4C – Pure Y1 page 67

25

2 Use a separate diagram to sketch each pair of graphs. **a** $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$